Project PRISM

Passive RF-Based Indoor Spatial Mapping

Real-Time Wi-Fi Zonal Localization using ESP32 CSI

Jayinaksha Vyas — 2302CM08
Adwik Verma — 2301EC54
Utsav Das — 2301EC52
Anurag Semwal — 2302CM02
Nishant Kumar— 2301EC53

What is CSI? Channel State Information

OFDM Subcarriers

Standard 802.11n Wi-Fi (2.4 GHz) uses OFDM — the spectrum is split into 64 individual sub-channels called subcarriers.

Amplitude + Phase

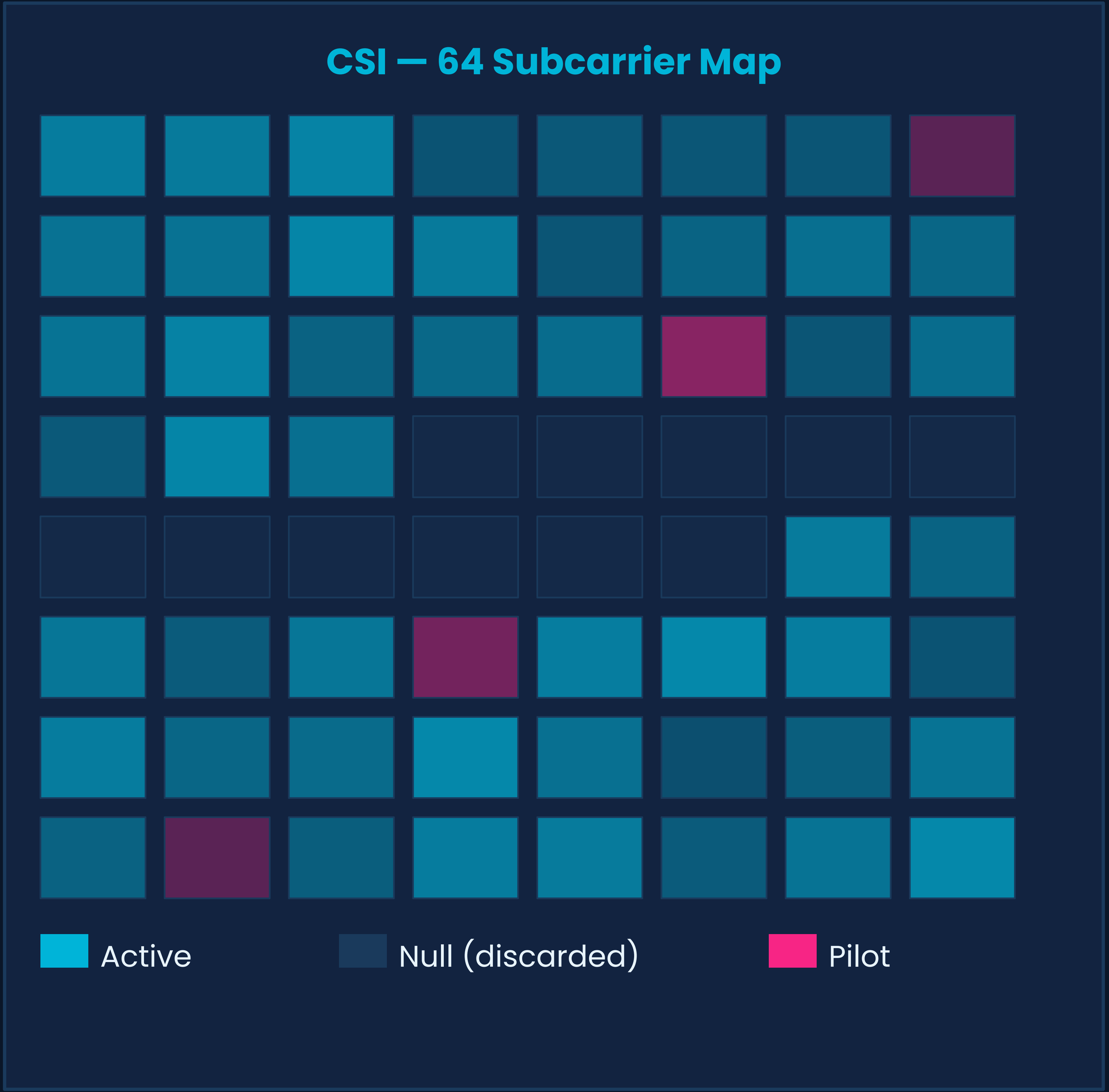
CSI gives the amplitude and phase of each subcarrier — a fingerprint of how the RF environment looks at that instant.

Human Body Effect

When a person moves, they disturb the multipath reflections, causing measurable Doppler shifts and amplitude fluctuations.

53 Active Streams

After discarding null/pilot subcarriers (27–37), we retain 53 active streams — each a live RF sensor.



System Architecture

ESP32 MCU
Wi-Fi Sniffer
(ESP-IDF firmware)

Serial USB
115200 baud
CSI stream

DSP Pipeline
Python
SciPy / NumPy

ML Engine
Random Forest
135 features

Hardware

- ESP32 NodeMCU (single antenna)
- 802.11n monitor mode
- 28-element complex CSI array

Signal Processing

- Hampel filter for spike removal
- Background subtraction (100-pkt avg)
- Butterworth bandpass 0.1–3.0 Hz

Classification

- 135-dimensional feature vector
- Random Forest / HistGradientBoost
- 50% confidence threshold gate

Raw Signal Extraction

128-Element Array

The ESP32 returns a 128-element complex array (64 subcarriers × I + Q components) per packet at ~100 packets/sec.

Phase Unusable

Single-antenna clock drift makes raw phase data completely unreliable — phase offsets shift arbitrarily across packets.

Amplitude Calculation

We compute the Euclidean magnitude per subcarrier:
$|H| = \sqrt{Real^2 + Imaginary^2}$
This gives a clean, drift-immune metric.

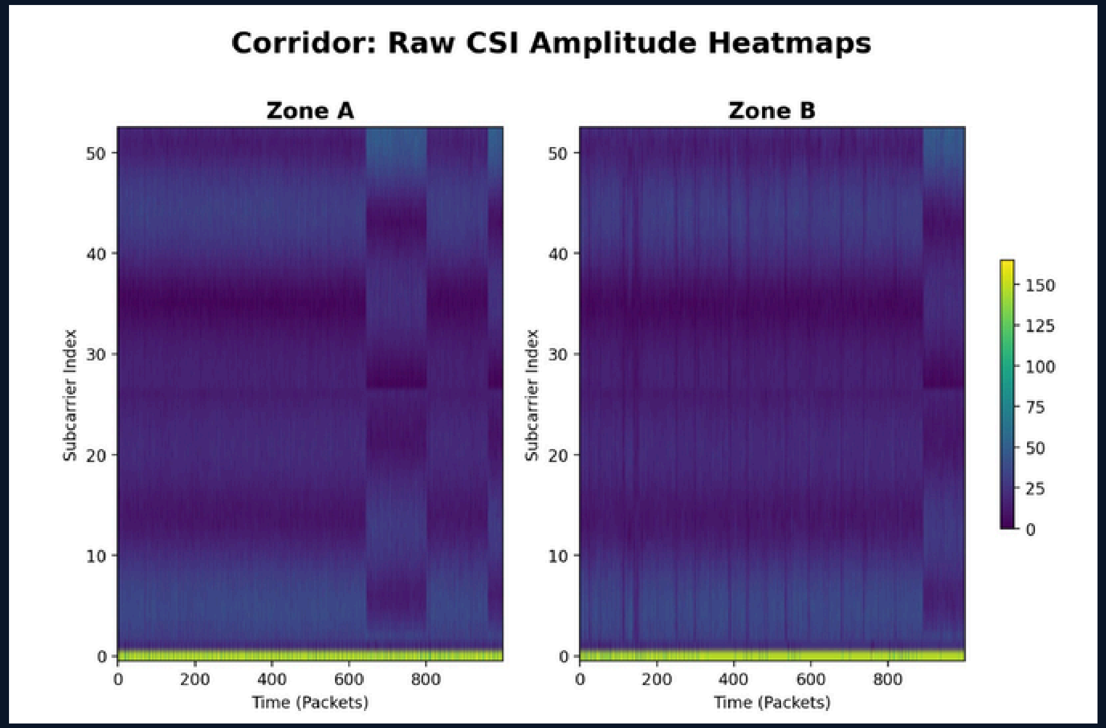
53 Active Streams

Null subcarriers (indices 27–37) and pilot tones are discarded, leaving 53 live RF amplitude streams.

Amplitude Formula

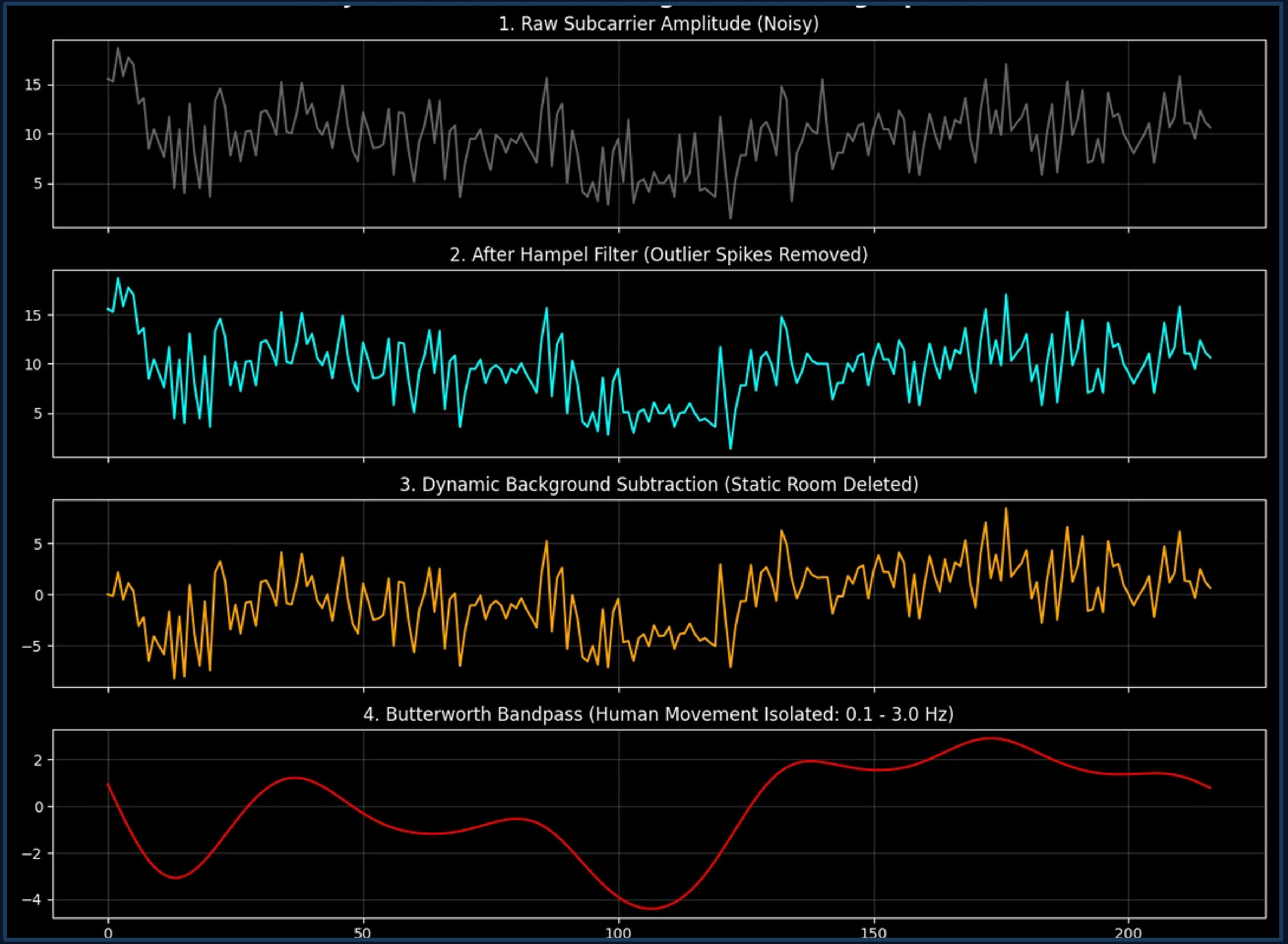
$$|H[k]| = \sqrt{Re^2 + Im^2}$$

Total subcarriers	64
Null / pilot (removed)	11
Active streams	53
Packet rate	~100 pkt/s
Array size (I+Q)	128 elements



DSP Pipeline — Stage 1: Noise Reduction

Problem: Environmental interference (Bluetooth, microwave ovens, cosmic rays) causes massive amplitude spikes that corrupt the human-motion signal.



How Hampel Filter Works

① Slide a window of N packets along each subcarrier stream

② Calculate the median and MAD (Median Absolute Deviation)

③ Flag any sample > 3 × MAD from the median as an outlier

④ Replace flagged spikes with the local median value

✓ Result: Smooth signal, zero data loss, preserves motion dynamics

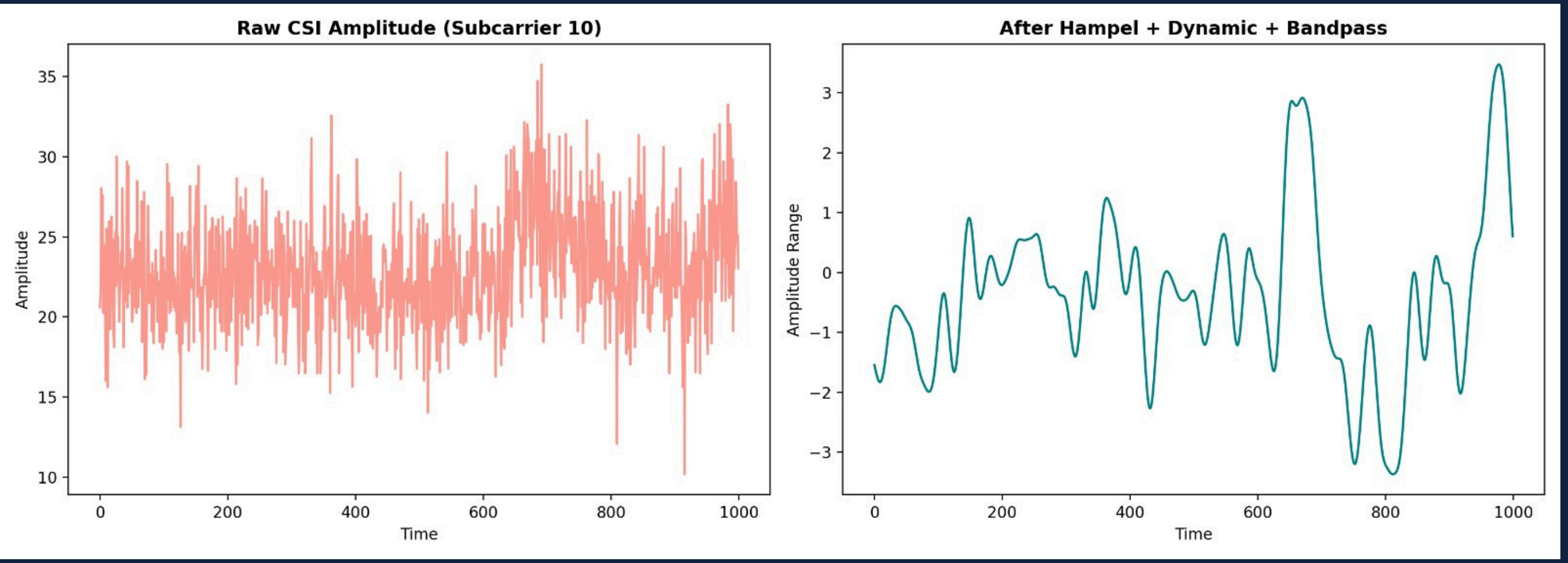
DSP Pipeline — Stages 2 & 3: Human Isolation

Stage 2 — Background Subtraction

The static room — walls, furniture, ceiling — dominates the raw CSI amplitude by 80–90%. Subtracting a trailing 100-packet moving average mathematically removes the room's geometry, leaving only the dynamic human-motion residual.

$$X_{motion}[t] = X[t] - \mathrm{mean}(X[t-100 : t])$$

Effect: Zero-centers the signal around human perturbations. Empty room → flat line at zero. Human present → visible oscillations.



Stage 3 — Butterworth Bandpass

Human physiological motion produces specific Doppler-shift frequencies. A 3rd-order Butterworth bandpass filter retains only the meaningful bio-motion band, rejecting low-frequency HVAC drift and high-frequency electronic noise.

Breathing (resting)	0.1 – 0.5 Hz
Breathing (active)	0.5 – 1.0 Hz
Walking cadence	1.0 – 3.0 Hz
HVAC / static drift	< 0.1 Hz
Electronic noise	> 3.0 Hz

Filter: 3rd-order Butterworth [0.1 – 3.0 Hz]

Data Collection Environments

Environment 1 — Corridor

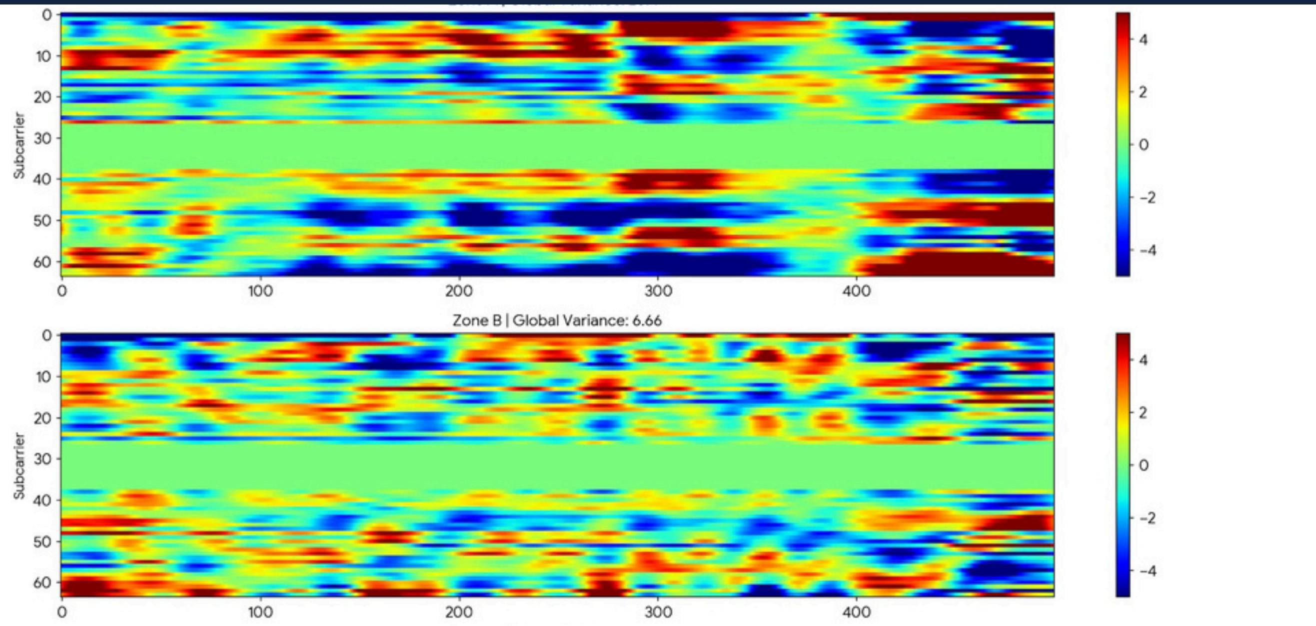
Zones: A & B (linear layout)

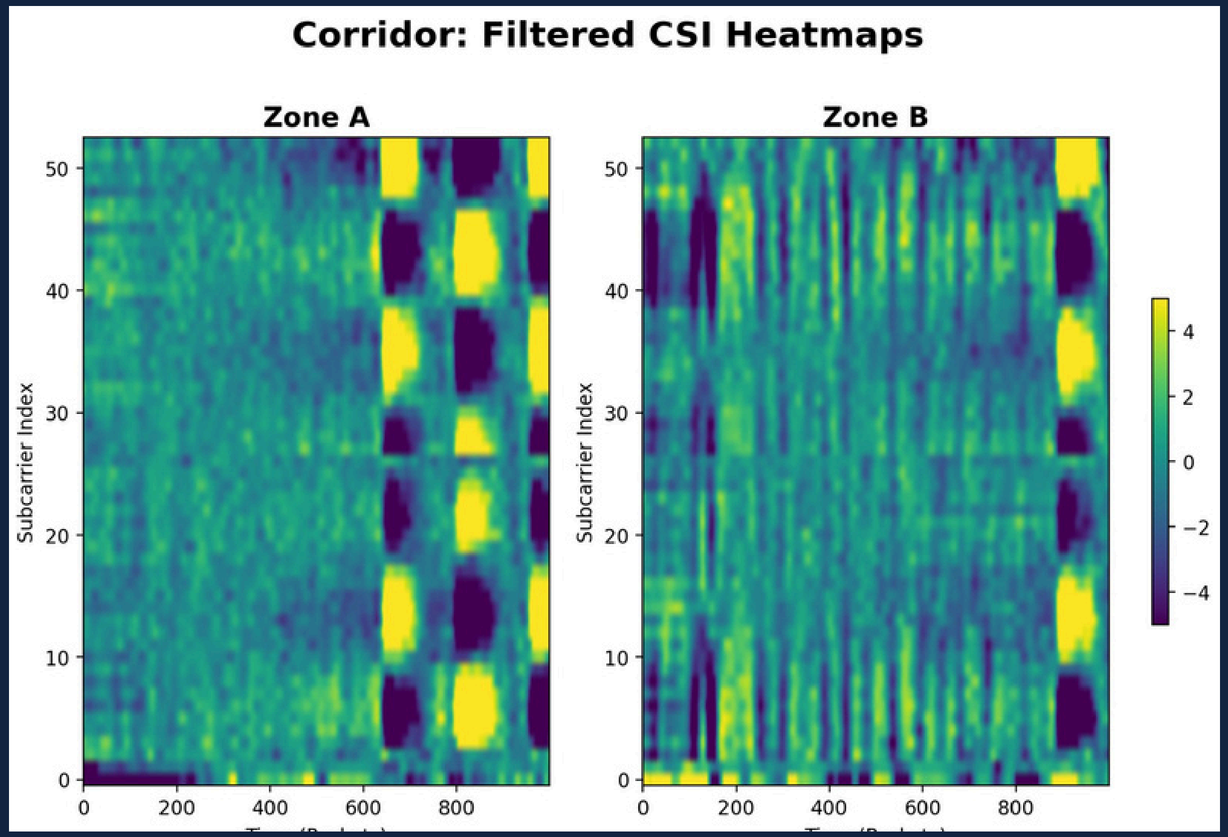
Challenge: Geometrically symmetric — both zones produced nearly identical multipath signatures, creating zone confusion for the classifier.

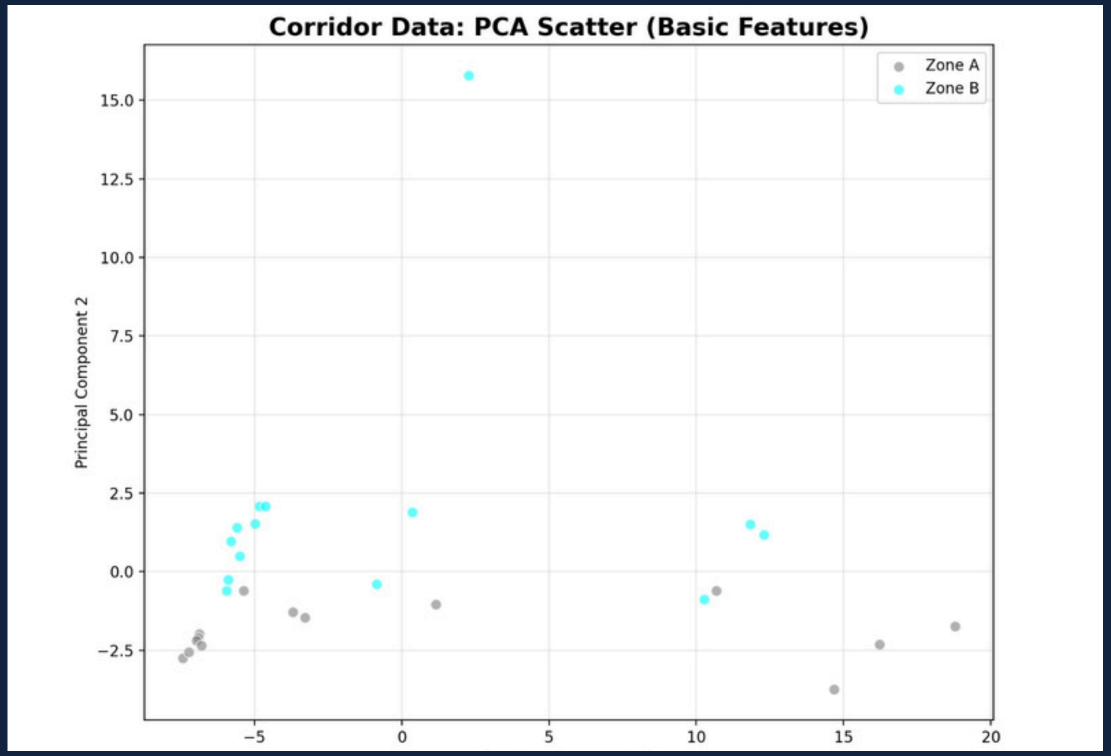
Result: Lower classification accuracy. Symmetrical layouts violate the assumption of geometrically distinct RF fingerprints.

Zone A	Zone B

↔ Symmetric multipath — hard to distinguish







Feature Engineering — The Dimensionality Upgrade

53
Raw subcarrier streams

135
Engineered feature dimensions

4
Zone classes

1s
Window duration

Per-Subcarrier Stats

Mean amplitude (53 values)

Variance (53 values)

→ *106 base features*

Multi-Lag Autocorrelation

Lags 1, 5, 10 applied per stream

Captures rhythmic vs. transient motion

→ *Key discriminator: breathing vs. walking*

Covariance Eigenvalues

53×53 subcarrier covariance matrix

Top-5 eigenvalues extracted

→ *Maps multipath fading complexity*

Temporal Non-Stationarity

Variance ratio: 1st-half vs 2nd-half window

Detects subjects crossing RF boundary

→ *Distinguishes static vs. moving subject*

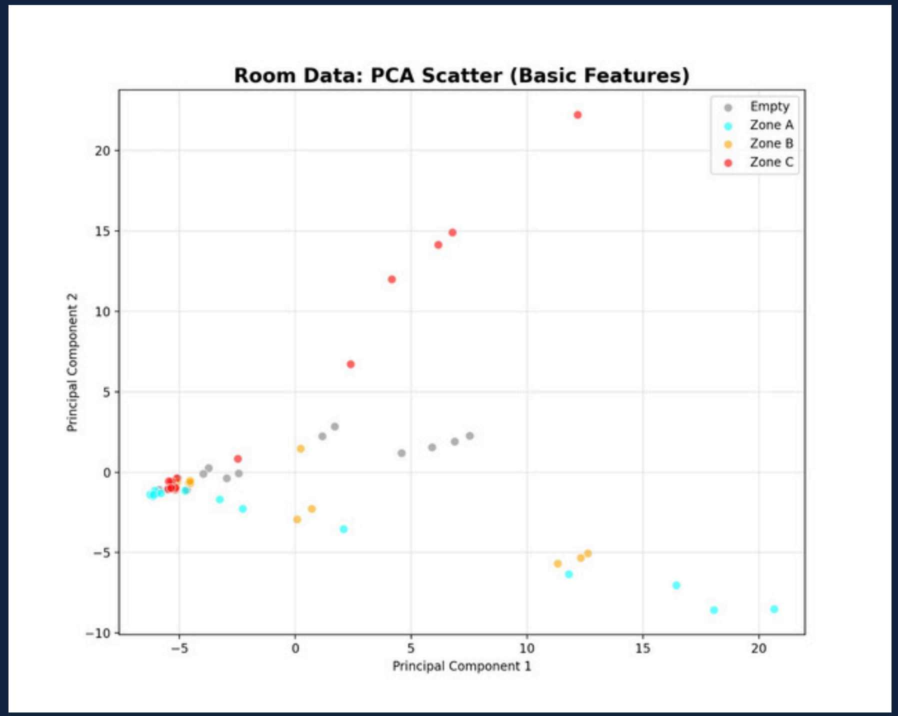
Advanced Spectral Features

Feature 1 — Temporal Non-Stationarity

```
Variance Ratio  =  Var(first 50 packets) / Var(last 50 packets)
```

Why it matters:
- Ratio ≈ 1.0 → Subject standing still in zone
- Ratio >> 1 or << 1 → Subject crossing the RF boundary between zones

This single feature dramatically improves zone-transition detection and prevents the classifier from confusing 'Zone A' with 'entering Zone B'.



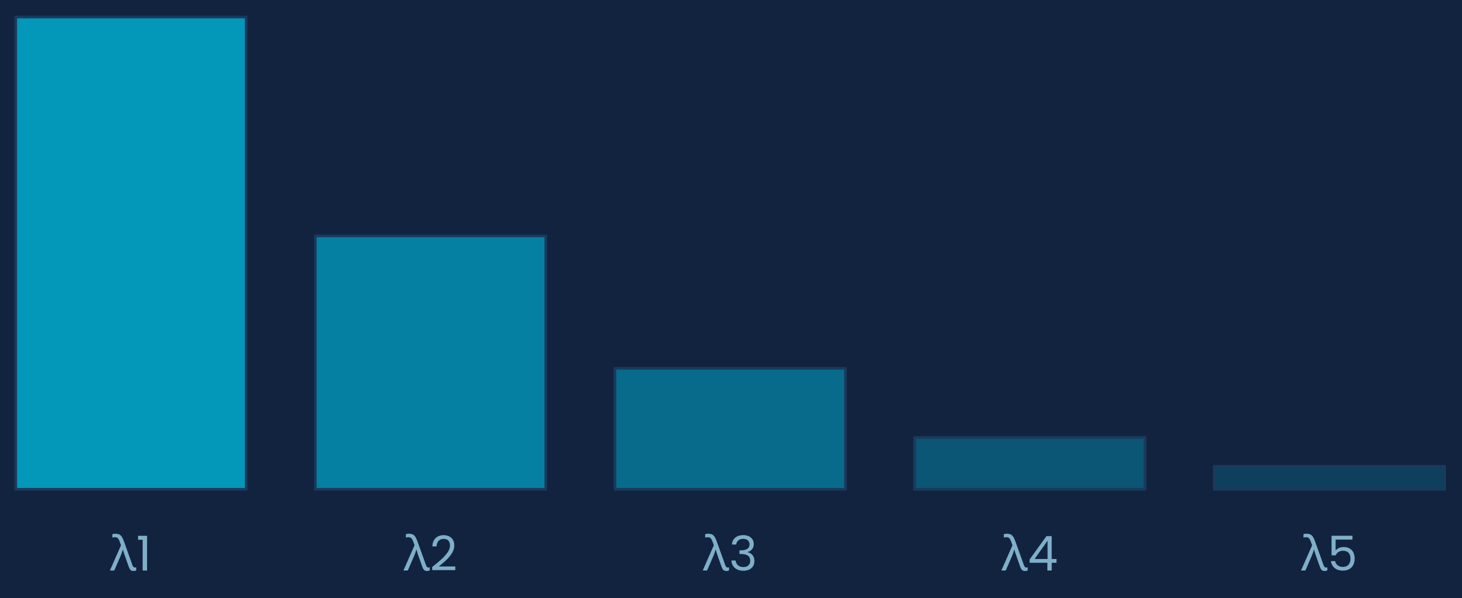
Feature 2 — Covariance Eigenvalues

```
Cov(X) → Top-5 Eigenvalues
```

Physical meaning:
- Large λ_1 → Single dominant reflection path
- Spread eigenvalues → Rich multipath environment

This maps how complex the RF reflections are in each zone. A corner room (Zone C) has many reflections from three walls → spread eigenvalues. Open center area (Zone A) has fewer → concentrated spectrum.

Eigenvalue Spectrum (example)



Machine Learning Model Selection

Why Random Forest beat Gradient Boosting

✗ Discarded: Support Vector Machine

High-Dimensional Correlation

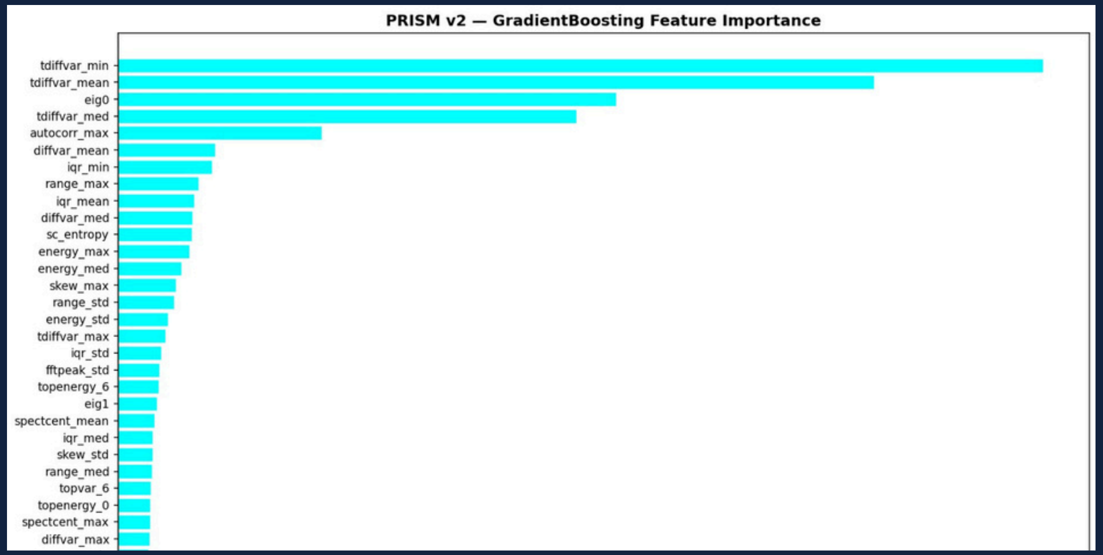
135 features with high inter-feature correlation make kernel choices brittle and overfitting likely.

StandardScaler Mismatch

Strict Standard Scaling equalizes all subcarrier magnitudes — destroying the physically meaningful relative amplitude differences between zones.

Kernel Sensitivity

RBF / polynomial kernels required extensive grid-search tuning, with no stable optimal found across different room geometries.



✓ Chosen: Random Forest Ensemble

Implicit Feature Selection

Decision trees internally rank and select the most discriminative subcarriers — noisy features are naturally ignored without explicit tuning.

Non-Linear Boundaries

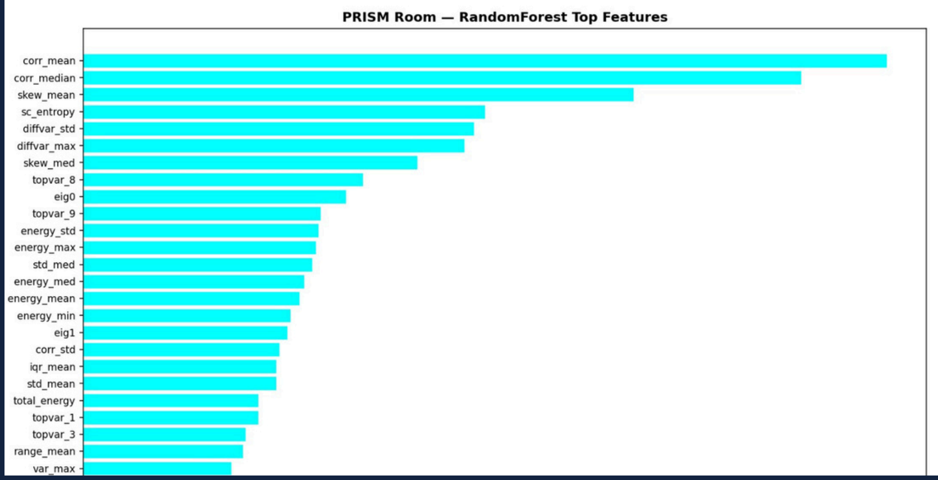
Handles non-linear zone boundaries in the 135D feature space that linear SVMs simply cannot model.

Robust to Outliers

Bagging across 100+ trees makes the ensemble robust to individual corrupted samples that Hampel filter may have missed.

No Scaling Required

Tree-based splits are scale-invariant — the physically meaningful amplitude magnitude differences are preserved.



Overcoming the Data Leakage Trap

⚠ **Initial Model: 96.3% Accuracy** → Failed completely in live inference. The number was fabricated by leakage.

The Bug

Sliding window with step=10 created 90% overlap between consecutive windows.

K-Fold split at the window level placed near-identical frames in both train and test sets.

The Cause

Temporal leakage: The classifier memorized local noise signatures from adjacent windows rather than learning the physical spatial features.

Live data = new room → complete failure.

The Fix

Increased step to 50 → 50% non-overlapping windows.

K-Fold now splits on packet sessions, not windows.

Result: Honest generalized accuracy measurement.

High cross-validation accuracy on time-series data is meaningless without ensuring temporal separation between train and test splits.

Honest Validation & Final Results

73.3%
Generalized
2-Zone Accuracy

>83%
Empty Zone
Recall

50%
Window
Overlap (fixed)

8×
Gaussian Noise
Augmentation

Corrected Validation Methodology

- Non-overlapping windows with step=50 (50% overlap)
- Session-level K-Fold split — no temporal leakage
- 8× augmentation: Gaussian noise scaled to per-subcarrier variance σ
- Test set: unseen room geometry (different day, repositioned ESP32)
- Empty zone prioritized — recall > 83% for safety-critical detection

Per-Zone Classification Accuracy

Live Inference Architecture

1

Ring Buffer Ingestion

Pre-allocated NumPy deque buffers eliminate Serial I/O stall. Continuous packet ingestion at ~100 Hz without blocking inference.

2

DSP Processing

Real-time Hampel → background subtraction → Butterworth applied to 100-packet windows as they fill.

3

Confidence Gating

Random Forest must output ≥ 50% probability for any class to trigger a zone update. Ambiguous predictions are silently discarded.

4

Exponential Vote Queue

3-vote exponential queue with 1.0s release timeout. Only unanimous (3/3) or supermajority votes flip the UI — preventing flicker.

UI Stabilisation Strategy

Stage 1 — Confidence Gate (RF probability ≥ 50%):

Blocks low-confidence predictions from ever reaching the display layer.

Stage 2 — Exponential Queue (3 votes + 1.0s timeout):

Requires sustained zone prediction before committing — eliminates sub-second flicker.

Conclusion

- Single-antenna ESP32 can perform 4-zone spatial mapping without cameras or wearables
- Heavy DSP pre-processing (Hampel + background sub + Butterworth) is mandatory — raw CSI is unusable
- 135-dimensional time-frequency features dramatically outperform naive amplitude statistics
- 73.3% generalized accuracy across zones; >83% recall on empty-room detection
- Temporal leakage is the #1 failure mode in time-series ML — corrected via session-level K-Fold

Thank You